

Project 2: Analysis Plan

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Aim 1

Analysis Plan

For Aim 1, we will evaluate whether peripheral inflammatory markers are associated with one-year change in episodic memory and AD-signature cortical thickness. We will assess whether (a) higher baseline cytokine and chemokine levels and (b) greater increases in these markers over time are associated with greater decline in these outcomes. Outcomes are change from baseline to year 1 in CVLT, Benson Figure delayed recall, Mnemonic Similarity Task, one-week delayed story recall, and mean cortical thickness across AD-signature regions. Primary predictors are IL-6, TNF- α , MCP-1, and Eotaxin-1.

Data will be screened for missingness, outliers, and collinearity using summary statistics and plots. Descriptive analyses will examine distributions of outcomes, predictors, and candidate covariates, along with correlations among inflammatory markers. Age, sex, and baseline outcome value will be included in all primary models. Additional covariates will be included if scientifically relevant and associated with predictors and/or outcomes, with sensitivity analyses assessing their impact. We will also compare the analytic sample with the full investigator-provided dataset on key characteristics. If collinearity among the inflammatory markers is acceptable, multivariable linear regression models including all four markers will be fit, and the primary hypothesis test will be the overall F-test for the inflammatory marker

block at $\alpha = 0.025$. Otherwise, separate adjusted linear regression models will be fit for each marker, and the primary test will be the t-test for the marker coefficient. For Hypothesis (a), baseline predictors will be related to one-year change outcomes; for Hypothesis (b), change in predictors from baseline to year 1 will be related to the same outcomes.

Power

Because enrollment is $N = 175$, power rather than sample size was evaluated. Power was estimated in the *powertools* package. For joint models, *powertools::mlrF.overall()* was used to estimate power for the overall F-test of the four inflammatory markers at $\alpha = 0.025$ across plausible R^2 values informed by preliminary data on baseline IL-6, baseline MCP-1, and one-year change in CVLT and cortical thickness. Observed R^2 values ranged from about 0.11 to 0.48 so analyses considered $R^2 = 0.05$ to 0.60. Because preliminary data were available only for baseline inflammatory markers and one-year change outcomes, the same effect-size scenarios were used to approximate power for both Aim 1 hypotheses. As shown in Table 1, power for joint models exceeded 90% at $R^2 = 0.10$ and was essentially 100% for larger effects. For separate-marker models, *powertools::corr.1samp* was used to estimate power at $\alpha = 0.025$ for correlations from 0.10 to 0.90. As shown in Table 2, power was low for small effects ($r = 0.10$, power = 0.18) but reached 0.87 at $r = 0.25$ and was near 100% for larger effects. Thus, the study is well powered for modest to large associations but may be underpowered for small effects. Because preliminary data were unavailable for all planned covariates, these calculations were based on simplified models; adjustment for precision variables may increase power, whereas adjustment for strong confounders may reduce it.

Table 1: Power for Aim 1 using joint models

R^2	Power
0.05	0.55
0.10	0.92
0.20	1.00
0.35	1.00
0.50	1.00
0.60	1.00

Table 2: Power for Aim 1 using separate models

ρ	Power
0.10	0.18
0.25	0.87
0.40	1.00
0.50	1.00
0.60	1.00
0.75	1.00
0.90	1.00

Aim 2

Analysis Plan

For Aim 2, we will evaluate whether peripheral inflammation modifies the association between amyloid burden and one-year change in cognition and cortical thickness. We hypothesize that individuals with both elevated inflammatory markers and high amyloid burden will experience the greatest declines over one year. For each inflammatory marker, adjusted linear regression models will include the marker, amyloid deposition, their interaction, baseline outcome value, age, sex, and other selected covariates. The primary test will be the marker-by-amyloid interaction coefficient at $\alpha = 0.025$. If collinearity among inflammatory markers is acceptable, models including all markers and their interaction terms will also be considered, with inference based on a joint F-test of the interaction block.

Power

For Aim 2, power was estimated using *powertools::corr.2samp()* to detect differences in the marker-outcome correlation across dichotomized amyloid groups. This provided a simplified approximation to the planned interaction analysis because preliminary data on amyloid deposition and additional covariates were unavailable. As shown in Table 3, power depended strongly on both the correlation difference and the balance of the amyloid groups. For small differences ($\Delta\rho = 0.1$), power was low across all splits (0.064 to 0.097). For $\Delta\rho = 0.2$, power remained limited (0.147 to 0.272). Power increased substantially for larger differences, exceeding 0.50 when $\Delta\rho = 0.4$ in more balanced splits and exceeding 0.90 when $\Delta\rho = 0.5$ for most splits. Overall, the study is reasonably powered to detect moderate to large effect modification but not small interaction effects, especially when amyloid groups are highly imbalanced.

Table 3: Power for Aim 2 across correlation differences and amyloid group sample-size splits

$\Delta\rho$	Amyloid Deposition Group Sample Size Splits					
	150/25	125/50	100/75	75/100	50/125	25/150
0.1	0.067	0.088	0.097	0.097	0.086	0.064
0.2	0.154	0.234	0.272	0.271	0.230	0.147
0.3	0.305	0.478	0.553	0.551	0.471	0.292
0.4	0.519	0.753	0.828	0.826	0.746	0.501
0.5	0.754	0.937	0.970	0.969	0.933	0.737

Type I Error Control

To account for multiple comparisons across outcomes, we controlled the family-wise Type I error rate at $\alpha = 0.05$ by defining two outcome domains: episodic memory outcomes and AD-signature cortical thickness. Because these domains represent distinct constructs, we allocated the Type I error rate equally across the two families using a Bonferroni approach, which resulted in a significance threshold of $\alpha = 0.025$ for each domain. Because outcomes

within each domain are expected to be correlated, no further adjustment will be applied, and inference will emphasize consistency of effects across measures.

Reproducible Research Information

<https://github.com/eromo96/BIOS6624.git>

Data Location: /Users/eduardo/Eduardo/UColorado PhD Bios/Spring 26/BIOS6624

Advanced Statistical Methods/BIOS6624/Project2/Data